Four families of projective general linear maniplexes

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1. The question

The final paragraph of Section 5 in Leemans and Toledo [LT] proposes four infinite families of regular rank-four maniplexes with automorphism group $\mathrm{PGL}_2(q)$. In each family one end group is the full group; the other is respectively S_4, A_5, $\mathrm{PSL}_2(q)$, or $\mathrm{PGL}_2(q)$. We construct all four families over prime fields. The question asks for infinitely many examples in each class, rather than an example for every prime power or a classification of all examples.

Throughout, p is an odd prime and

$$G = \mathrm{PGL}_2(\mathbf{F}_p), \qquad T = \mathrm{PSL}_2(\mathbf{F}_p).$$

A string marking of rank four is a generating tuple (a, b, c, d) of pairwise distinct nonidentity involutions satisfying

$$[a, c] = [a, d] = [b, d] = 1. \tag{1}$$

Its two end groups are $\langle a, b, c\rangle$ and $\langle b, c, d\rangle$. We use the correspondence between such markings and regular maniplexes from [LT, Section 2]. For the needed direction the proof is immediate: take the flags to be G and adjacency of colour i to be left multiplication by the corresponding marker. Every adjacency and every product of two different adjacencies is fixed-point-free, since the markers are nonidentity and distinct. Equation (1) gives the required commutations, and generation gives connectedness. An automorphism preserving colours commutes with all left translations. Its value at the identity determines it as a right translation. Thus the full group of colour-preserving automorphisms is the right regular action of G, and the maniplex is regular. This argument imposes no intersection property.

Theorem 1. For every prime in each of the following rows there is a string marking of G with the indicated end groups.

Prime condition	$\langle a, b, c\rangle$	$\langle b, c, d\rangle$
$p > 5$, $p \equiv 3, 5 \pmod 8$	S_4	G
$p \equiv 9, 11 \pmod{20}$	A_5	G
$p \equiv 5 \pmod{12}$	T	G
$p > 3$, $p \equiv 3 \pmod 4$	G	G

Reversing the marking in the first three rows puts the full group at the facet end, as in [LT]. Dirichlet's theorem gives infinitely many primes in each indicated progression. Since $|G| = p(p^2-1)$ strictly increases, the resulting maniplexes are pairwise nonisomorphic. Hence Theorem 1 answers the four-family conjecture in the bound May 15, 2023 author edition of [LT]. A bounded search for later solutions found none; an exhaustive priority claim is not made.

2. Inputs from the common theory and from subgroup classification

The determinant modulo squares gives an epimorphism

$$\epsilon : G \longrightarrow \mathbf{F}_p^*/(\mathbf{F}_p^*)^2 \cong C_2, \qquad \ker \epsilon = T. \tag{2}$$

The common projective-line paper [E, Theorem 5] proves the following centralizer statement without any intersection hypothesis. If two projective involutions have product of order greater than two in odd characteristic, their common centralizer is trivial when the product is unipotent, and has order two when it is semisimple. If a matrix A represents the product in the latter case, the nonidentity common centralizer element is represented by

$$A - \frac{\operatorname{tr}(A)}{2} I. \tag{3}$$

We use this proved interface, rather than applying a theorem about polytopes to a maniplex that may fail intersection.

We retain the following precise classical classification inputs. By Giudici [G, Theorem 3.5, p. 10], for prime $p > 3$ a maximal subgroup of G not containing T is an affine point stabilizer, a dihedral torus normalizer, or an S_4 subgroup. The last possibility occurs exactly when $p \equiv \pm 3 \pmod 8$. The minor exceptional omission among the dihedral groups at $p = 5$ does not affect our use. There are no subfield cases when the field is prime. By [G, Theorem 2.2(6), p. 4], T has an A_5 subgroup for primes $p \equiv \pm 1 \pmod{10}$. These statements replace the unrestricted sentence about proper nonabelian subgroups in [LT, Section 5], which would omit affine, dihedral, and subfield groups. Both cited page numbers are also the physical page numbers of the bound Giudici PDF.

We also retain the usual finite Coxeter identifications $[3,3,3] = W(A_4)$ and $[3,3,4] = W(B_4)$, of orders 120 and 384. The first acts by adjacent transpositions on five letters. The second acts by adjacent coordinate swaps and a final sign change on four coordinates, giving $2^4 \cdot 4!$ elements. Their standard Coxeter presentations, rather than merely the existence of permutation quotients of those orders, are the retained input. Independent GAP calculations verify the exact orders of those literal presentations. The elementary quadratic-character formulas and Dirichlet's theorem are also classical inputs.

3. A constant quadruple of matrices

Over $\mathbf{F}_p$, with $p > 3$, put

$$R_0 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad R_1 = \begin{pmatrix} 1 & 0 \\ 1 & -1 \end{pmatrix}, \quad R_2 = \begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix}, \quad R_3 = \begin{pmatrix} 1 & -2 \\ 2 & -1 \end{pmatrix}. \tag{4}$$

Let a, b, c, d be their projective classes. Direct multiplication gives $R_i^2 = I$ for $i = 0, 1, 2$, and $R_3^2 = -3I$. Their determinants are

$$\det R_0 = \det R_1 = \det R_2 = -1, \qquad \det R_3 = 3. \tag{5}$$

Each matrix is nonscalar and invertible. The three nonadjacent pairs anticommute as matrices, so their projective classes satisfy (1). The classes are pairwise distinct: the zero patterns distinguish R_0, R_1, R_2, whereas R_3 has no zero entry. Thus the four classes are distinct nonidentity involutions. Furthermore,

$$R_1 R_2 = -\begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix}, \qquad (R_0 R_1)^3 = -I. \tag{6}$$

The second product is nonscalar, so $|ab| = 3$. If $L(t) = \left(\begin{smallmatrix} 1 & 0 \\ t & 1 \end{smallmatrix}\right)$, then $L(1)^n = L(n)$. It is scalar exactly when $n = 0$ in the field. Consequently $|bc| = p$, and its powers run through the full group $U = \{[L(t)] : t \in \mathbf{F}_p\}$, since the field is prime.

We act on the projective line by fractional linear transformations. The unique common fixed point of U is 0. Both a and d move this point: $a(0) = \infty$ and $d(0) = 2$. Each end group therefore satisfies the following elementary criterion.

Lemma 2. If $H \leq G$ contains U and an element g with $g(0) \neq 0$, then $T \leq H$.

Proof. The group gUg^{-1} is the full unipotent subgroup fixing $g(0)$. The action of U on the complement of 0 is transitive: reciprocation identifies it with addition on $\mathbf{F}_p$. Choose $u \in U$ carrying $g(0)$ to ∞. Then H contains $ugUg^{-1}u^{-1}$, the full upper unipotent subgroup.

Upper and lower elementary matrices generate SL_2. Here is the needed elimination argument. Write $V(t) = \left(\begin{smallmatrix} 1 & t \\ 0 & 1 \end{smallmatrix}\right)$. For $t \neq 0$,

$$W(t) = V(t)L(-t^{-1})V(t) = \begin{pmatrix} 0 & t \\ -t^{-1} & 0 \end{pmatrix}, \qquad W(t)W(-1) = \mathrm{diag}(t, t^{-1}).$$

Multiplication by $W(1)$ makes the top left entry of a determinant-one matrix nonzero if necessary. A lower elementary operation then clears the bottom left entry, an upper operation clears the top right entry, and the remaining diagonal matrix has the displayed form. Thus every determinant-one matrix is a product of these elementary matrices. Passing to projective classes proves the lemma. □

Both end groups of (4) contain T. If $p \equiv 3 \pmod 4$, then -1 is nonsquare, so each end contains $b \notin T$ and equals G. This proves the fourth row of Theorem 1. If $p \equiv 5 \pmod{12}$, then -1 is square and 3 is nonsquare. Equations (2) and (5) give $\langle a, b, c\rangle = T$ and $\langle b, c, d\rangle = G$, proving the third row, including $p = 5$. Neither argument needs the exceptional subgroup classification.

This quadruple also explains its relationship to [LT]. When $p \equiv 1 \pmod{12}$, both determinants are square, and both ends are T. Choose s with $s^2 = -1$ and put $D = \mathrm{diag}(-s, 1)$. If S_i are the first three matrices of [LT, Lemma 4.5], then $DS_0D^{-1} = -sR_0$, $DS_1D^{-1} = sR_1$, and $DS_2D^{-1} = sR_2$. Their projective classes are therefore the first three classes of (4). The fourth is the unique nonidentity common centralizer of the first two, so it agrees as well. Thus (4) is a

normalization of the source's known family. The contribution here is its use at other determinant classes, with an elementary argument controlling both end groups.

4. Excluding affine and dihedral end groups

The remaining constructions extend an exceptional local group by the centralizer element (3). We first isolate the obstruction needed to show that the opposite end is the full group.

Lemma 3. Suppose (a, b, c, d) is a string marking of G, with $|bc| = 3$ and $|ab| > 2$. Then $H = \langle b, c, d\rangle$ is contained in neither an affine point stabilizer nor a dihedral subgroup of G.

Proof. The centre of G is trivial. Indeed, a central projectivity must fix the unique common fixed point of each unipotent subgroup, so fixes every point of the projective line and is the identity.

In an affine point stabilizer in odd characteristic, a nonidentity involution has the form $x \mapsto -x+t$. Two such involutions commute only if they coincide, as their two products are translations by $t - u$ and $u - t$. The distinct commuting involutions b, d exclude this possibility.

Suppose next that H is dihedral. Since $|bc| = 3$, both b and c are noncentral reflections. If d were central in H, it would commute with b, c and also with a, making it central in G. Otherwise d is another reflection. As b, d are distinct and commute, $z = bd$ is the nonidentity central involution of H, and $d = bz$. Thus $cd = (cb)z$ has order six and $(cd)^3 = z$. In particular, $b = dz \in \langle c, d\rangle$. But a commutes with c, d, so would commute with b, contrary to $|ab| > 2$. A nonabelian subgroup of a dihedral group is itself dihedral, which proves the assertion about containment as well. □

5. The symmetric local group

Let $p > 5$ and $p \equiv \pm 3 \pmod 8$. The retained subgroup theorem gives a maximal subgroup $S \cong S_4$ of G not containing T. It cannot lie inside T, since then $S < T < G$ would contradict maximality. The restriction of ϵ to S is consequently its unique nontrivial homomorphism to C_2, namely the sign character.

Under an isomorphism with S_4, take

$$a = (1\,2), \qquad b = (2\,3), \qquad c = (3\,4). \tag{7}$$

These generate S, have $|ab| = |bc| = 3$, and satisfy $[a, c] = 1$. Since $p \neq 3$, the product ab is semisimple. By (3) the common centralizer of a, b in G is $\{1, d\}$. Inside S_4 the centralizer of $\langle (1\,2), (2\,3)\rangle$ is trivial: its centralizer on the first three letters is trivial, and the fourth letter must be fixed. Hence $d \notin S$. It is an involution, distinct from a, b, c, and maximality gives $\langle S, d\rangle = G$. All relations (1) hold.

It remains to prove $H = \langle b, c, d\rangle = G$. Since $b \notin T$, a proper H lies in a maximal subgroup not containing T: the only subgroups containing the index-two group T are T and G. The retained list and Lemma 3 would therefore put H inside an S_4 subgroup S'. Its intersection with T is A_4. Both b, c are odd involutions in S', hence transpositions, and $|bc| = 3$.

Relabel the four letters of S' so that $b = (1\,2)$ and $c = (2\,3)$. A distinct involution commuting with b is either $(3\,4)$ or $(1\,2)(3\,4)$. In the first case $|cd| = 3$; in the second $|cd| = 4$. Thus the full group $G = \langle a, b, c, d\rangle$ would be a quotient of $[3, 3, 3]$ or $[3, 3, 4]$. Its order would divide 120 or

384. This is impossible: the prime $p > 5$ divides $|G|$ and divides neither number. Hence $H = G$, proving the first row.

6. The alternating local group

Let $p \equiv 9, 11 \pmod{20}$. Since $p \equiv \pm 1 \pmod{10}$, the retained theorem supplies $A \cong A_5$ inside T. Choose

$$a = (1\,2)(3\,4), \qquad b = (2\,4)(3\,5), \qquad c = (1\,3)(2\,4) \tag{8}$$

under an isomorphism with A_5. These are distinct involutions with $[a, c] = 1$, $|ab| = 5$ and $|bc| = 3$. Their subgroup has order divisible by $2, 3, 5$, and an index-two subgroup of A_5 is impossible because A_5 is simple. They therefore generate A. This is the classical hemi-dodecahedral marking, retained as a local map.

The product ab is semisimple. Its centralizer is a cyclic torus of order $p - 1$ when $p \equiv 1 \pmod{5}$, and of order $p + 1$ when $p \equiv -1 \pmod{5}$. To see the determinant condition explicitly, the split torus is the group of eigenvalue ratios $\mathbf{F}_p^*$, on which ϵ is the ratio modulo squares. The nonsplit torus is $\mathbf{F}_{p^2}^*/\mathbf{F}_p^*$, acting by multiplication on the two-dimensional space $\mathbf{F}_{p^2}$. Its determinant is the field norm. The norm of a generator of $\mathbf{F}_{p^2}^*$ is a generator of $\mathbf{F}_p^*$, so ϵ is onto in this case too. Consequently the torus intersection with T has exactly half its order in either case.

The unique nonidentity element d centralizing both a, b is the involution of this torus. Indeed, conjugation by a inverts ab and acts by inversion on the torus, so precisely its identity and involution centralize a. If $p \equiv 11 \pmod{20}$, the order of the torus intersection with T is $(p-1)/2$, which is odd. If $p \equiv 9 \pmod{20}$, that order is $(p + 1)/2$, also odd. Thus $d \notin T$ in both cases.

Now $\langle A, d\rangle$ is not contained in T. None of the other proper maximal subgroups in the retained list contains A_5: the affine and dihedral groups are soluble, and S_4 has smaller order. Hence $\langle a, b, c, d\rangle = G$. The markers are distinct because $d \notin T$ and $a, b, c \in T$, and (1) follows from the choice of d.

Finally $H = \langle b, c, d\rangle$ is not contained in T. Lemma 3 excludes its containment in an affine or dihedral subgroup. If it were contained in $S' \cong S_4$, then $S' \cap T$ would be A_4. The involutions b, c would both lie in its normal Klein four subgroup and hence commute, contradicting $|bc| = 3$. The maximal-subgroup list therefore forces $H = G$, proving the second row. This completes the proof of Theorem 1. □

7. Attribution and verification

The conjecture, the use of local maps, the centralizer extension, and the underlying special-linear matrix family come from [LT]. The centralizer calculation is reused from [E] and has classical antecedents. The additions in this paper are the prime progressions selected by the determinant character, the extension of the normalized matrix family to two general-linear configurations, and the complete arguments forcing the opposite end to be the full group in the two exceptional cases. The subgroup classification, finite Coxeter presentations, quadratic-character formulas, and Dirichlet's theorem remain explicitly retained inputs. These attributions distinguish the new arguments of this project from claims of historical priority.

The author's GAP program constructs 16 actual marked groups, four in each family, checks the distant commutations and generation, and computes both end-group orders. An independent implementation constructs 17 examples using a different projective action, checks the complete small S_4 case split, and verifies the two Coxeter presentation orders. These calculations test the constructions; the written proofs above supply the universal prime quantifiers.

The accompanying Lean file has eleven declarations. They verify the matrix squares, determinants, anticommutations, adjacent products, nonscalar marks and pairwise nonproportionality, together with the exact lower-unipotent power formula, projective period and exhaustion of the prime-field root group. All use standard Lean axioms only. They do not formalize Lemma 2, the subgroup classification, the exceptional-group arguments, Dirichlet's theorem, or the complete four-family theorem. The runner checks the declared software versions and rejects altered matrices, collapsed markings, and a false unipotent period statement.

The source ledger fixes the exact 2023 edition and each retained interface. The independent review and its bounded priority search are separate records. This manuscript supplies one answer to the single four-family conjecture; the four cases do not count as four independent question results or as four recovered historical papers. No compression ratio is inferred from solving a question that the source posed.

References

[E] *Dihedral centralizers and regular polytopes over a projective line*, `phase230_rank_one_classification.md`, Theorem 5. Its exact frozen manifest and supporting matrix certificate are bound in this packet.

[G] Michael Giudici, *Maximal subgroups of almost simple groups with socle PSL(2,q)*, arXiv:math/0703685 (2007), Theorems 2.2 and 3.5. https://arxiv.org/abs/math/0703685.

[LT] Dimitri Leemans and Micael Toledo, *Maniplexes with automorphism group PSL(2,q)*, author manuscript dated May 15, 2023, Section 5, physical and printed page 9. Journal reference: Discrete Mathematics 346 (2023), article 113527, DOI 10.1016/j.disc.2023.113527. The present argument is bound to the author edition, not to an uncollated journal text. https://dipot.ulb.ac.be/dspace/bitstream/2013/359689/3/projectivemaniplexes_final.pdf.